Disease Risk Prediction using Machine Learning Classification System

Problem Id: 1

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SECTION – P(G2)

● **Problem Description:**

In modern healthcare systems, large volumes of patient clinical data are generated daily through hospitals, diagnostic centers, and wearable devices. However, extracting meaningful insights from this data to predict disease risk remains a challenging task.

Early prediction of diseases such as diabetes, heart disease, and hypertension can help in timely medical intervention, reduce treatment costs, and improve patient survival rates. Traditional diagnosis methods rely heavily on manual analysis, which can be time-consuming and prone to human error.

This project focuses on designing and developing a **complete end-to-end machine learning classification system** that predicts disease risk using patient clinical data. The system will use structured datasets containing features such as:

- Age
- Blood Pressure
- Glucose Level
- Cholesterol
- BMI (Body Mass Index)
- Lifestyle factors

The dataset will be processed through multiple stages including **data cleaning, normalization, and transformation** to ensure data quality and consistency. Feature selection methods will be applied to identify the most important variables affecting disease prediction.

The system will implement and compare multiple classification algorithms:

- Logistic Regression
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)

Finally, the models will be evaluated using standard performance metrics such as **accuracy, precision, recall, F1-score, confusion matrix, and ROC curve**.

This project demonstrates a **complete machine learning pipeline**, following best practices from data preprocessing to model evaluation and deployment readiness.

● **Objectives:**

➤ To build a reliable disease prediction system using ML algorithms
➤ To preprocess and clean real-world clinical data
➤ To identify important health-related features
➤ To compare multiple classification models
➤ To evaluate model performance using standard metrics

● **Steps Of Implementation:**

1. Dataset Collection:

- Select a clinical dataset (e.g., Heart Disease, Diabetes Dataset from Kaggle)
- Understand dataset structure and features

2. Data Preprocessing:

- Handle missing values (mean/median imputation)
- Remove duplicate or inconsistent data
- Normalize numerical features (Min-Max Scaling / Standardization)
- Encode categorical variables (Label Encoding / One-Hot Encoding)

3. Exploratory Data Analysis (EDA):

- Visualize feature distributions
- Identify correlations between variables
- Detect outliers

4. Feature Selection:

- Use correlation matrix
- Apply feature importance techniques
- Reduce irrelevant features to improve model performance

5. Model Selection:

- Logistic Regression (linear model)
- KNN (distance-based model)
- SVM (margin-based classifier)

6. Model Training:

- Split dataset into training and testing sets
- Train models using training data
- Tune hyperparameters

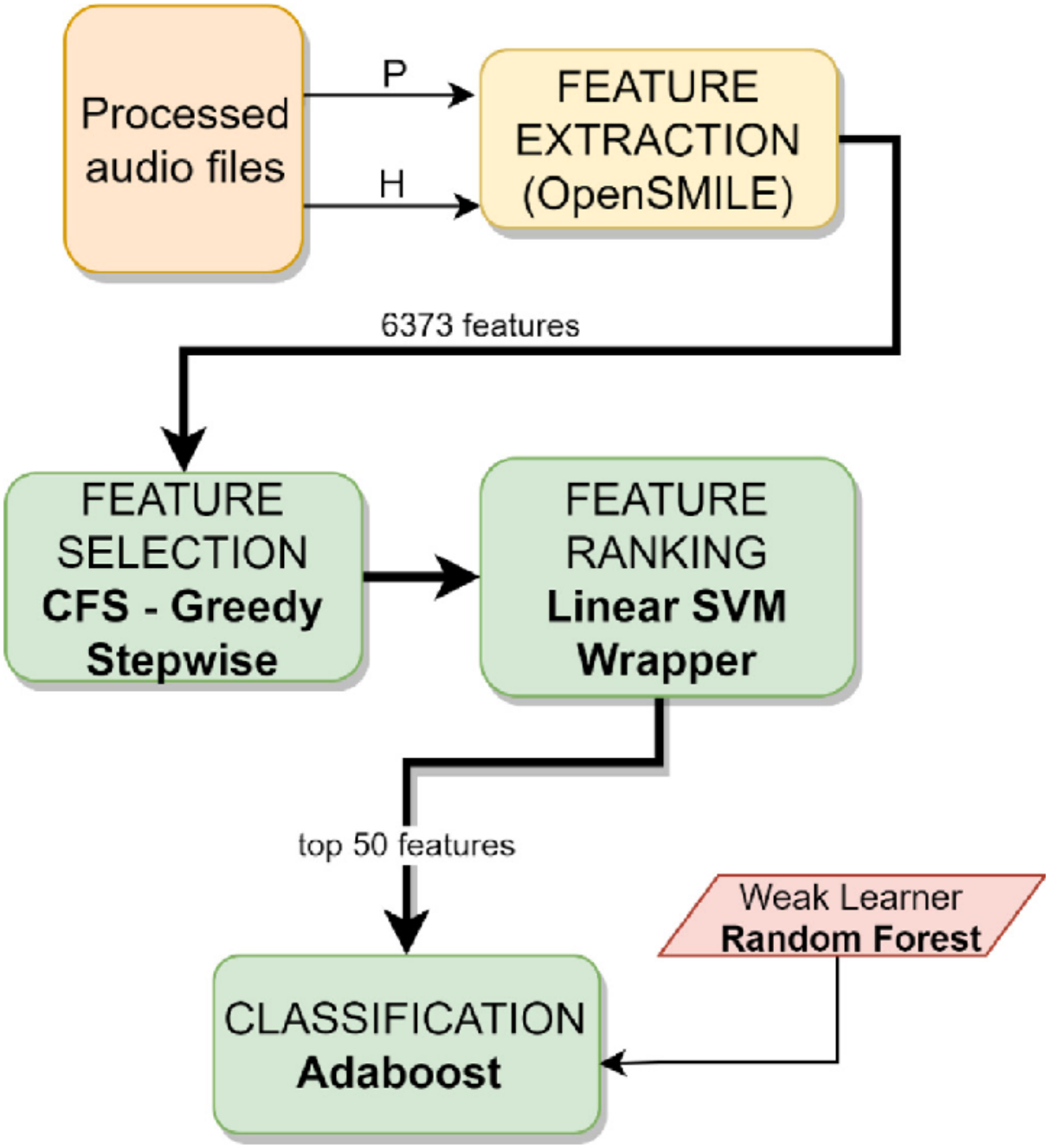
7. Model Evaluation:

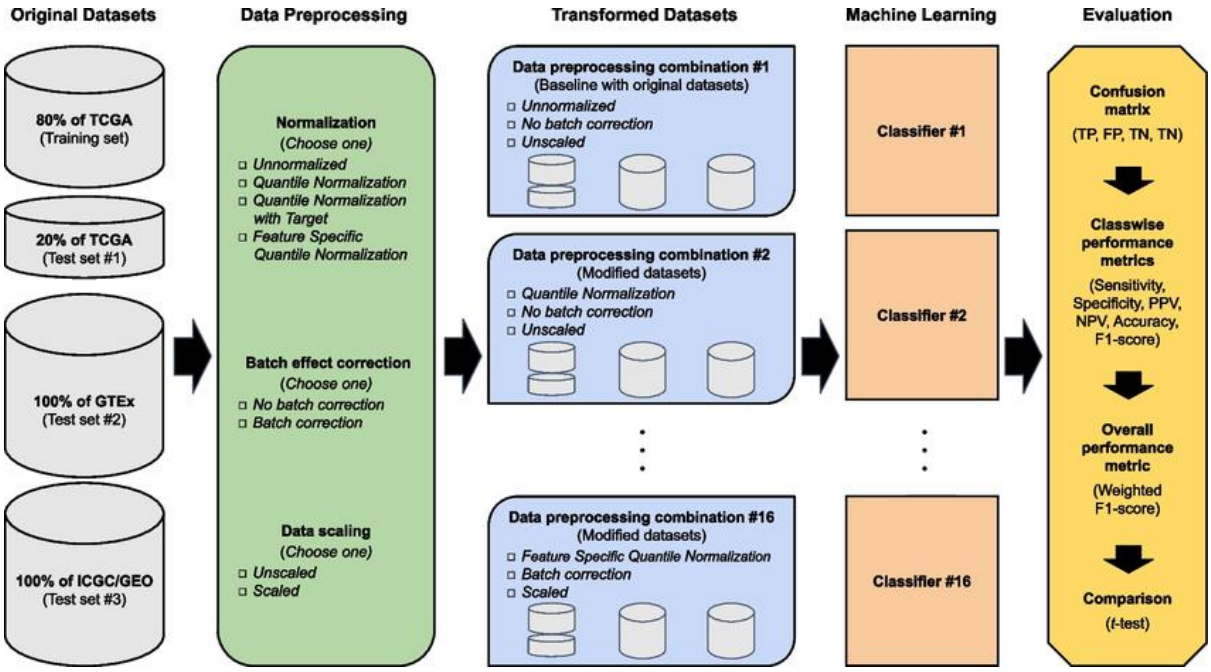
- Evaluate models using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
- Generate confusion matrix
- Plot ROC curve

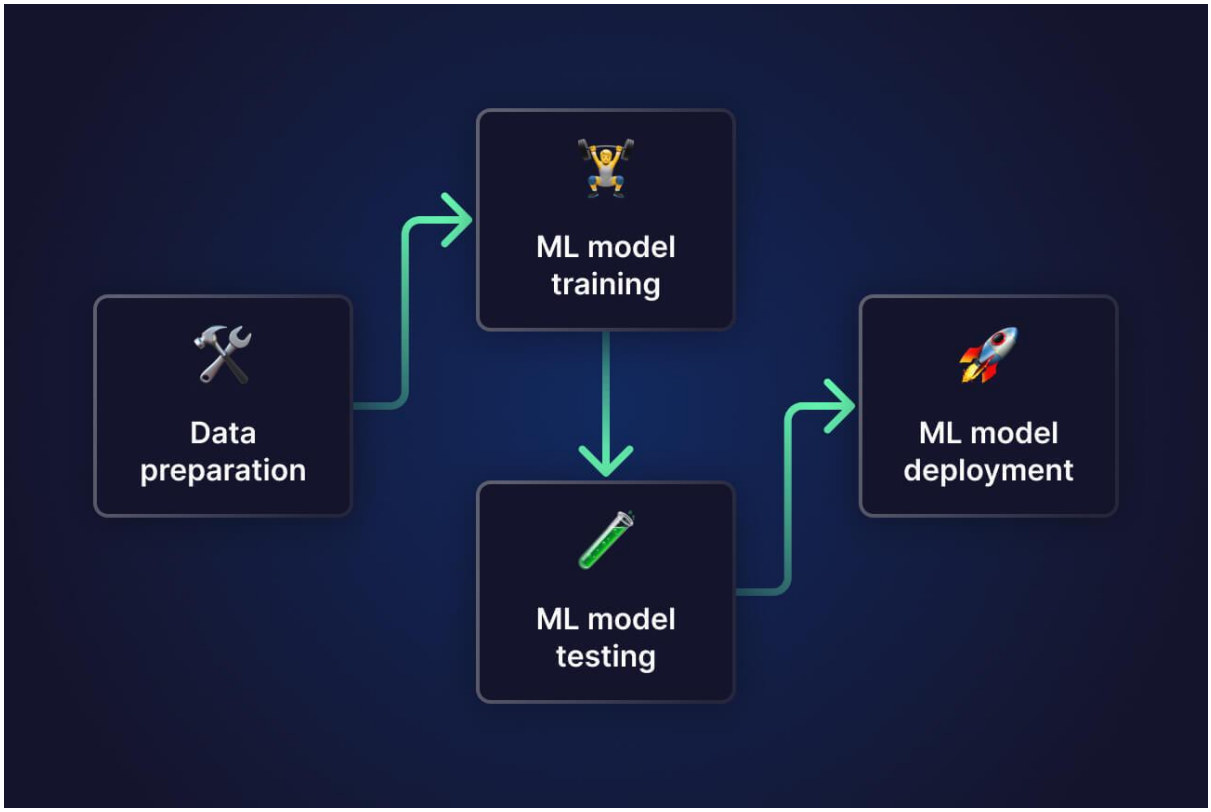
8. Result Analysis:

- Compare performance of all models
- Select best-performing model
- Interpret results

● **Flowchart of System (Main Pipeline):**


Processed audio files
P
H
FEATURE EXTRACTION (OpenSMILE)
6373 features
FEATURE SELECTION
CFS - Greedy Stepwise
FEATURE RANKING
Linear SVM Wrapper
top 50 features
Weak Learner
Random Forest
CLASSIFICATION
Adaboost





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Flow Explanation:

Patient Data → Data Cleaning → Feature Selection → Model Training → Model Evaluation → Best Model Output

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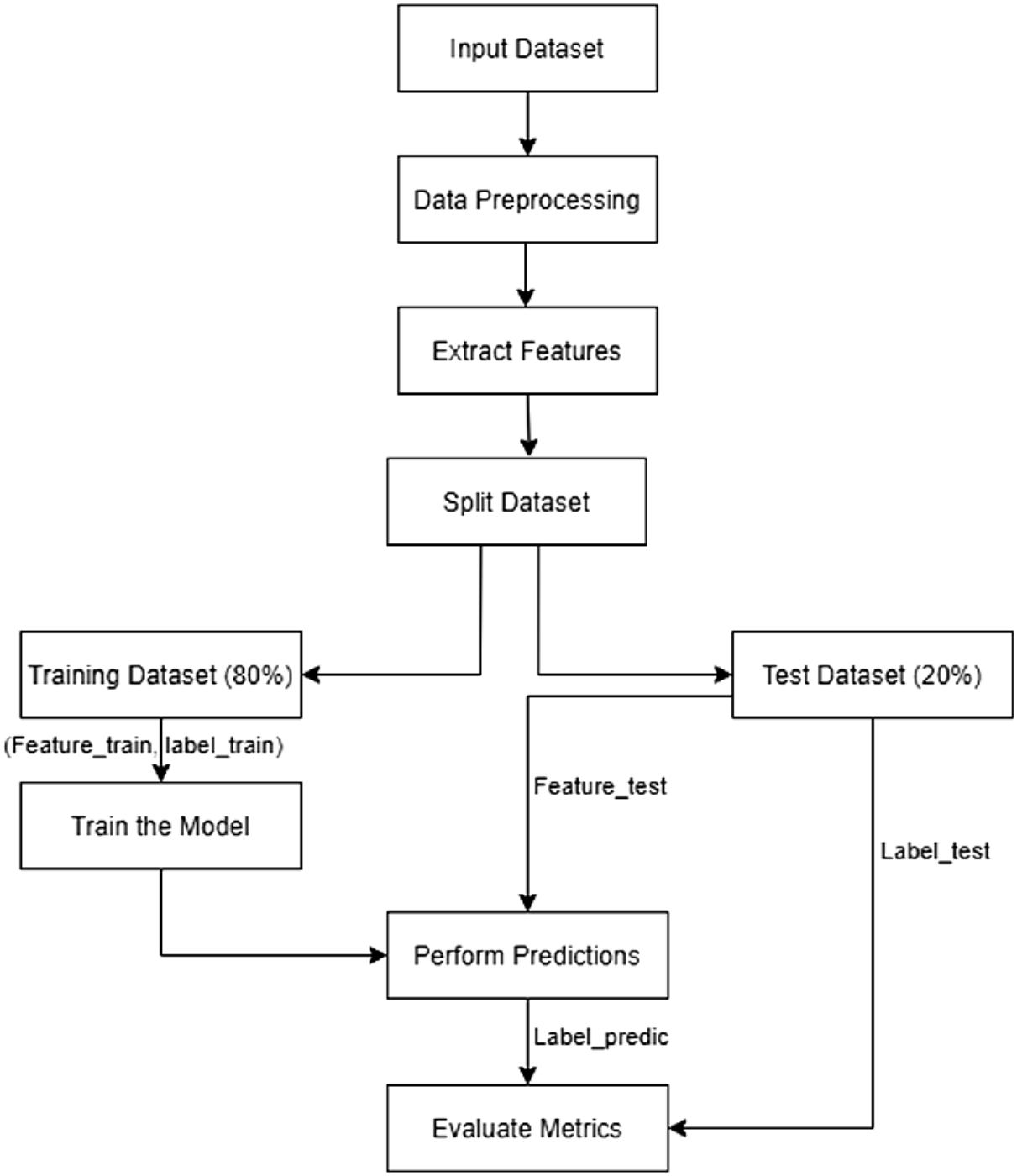
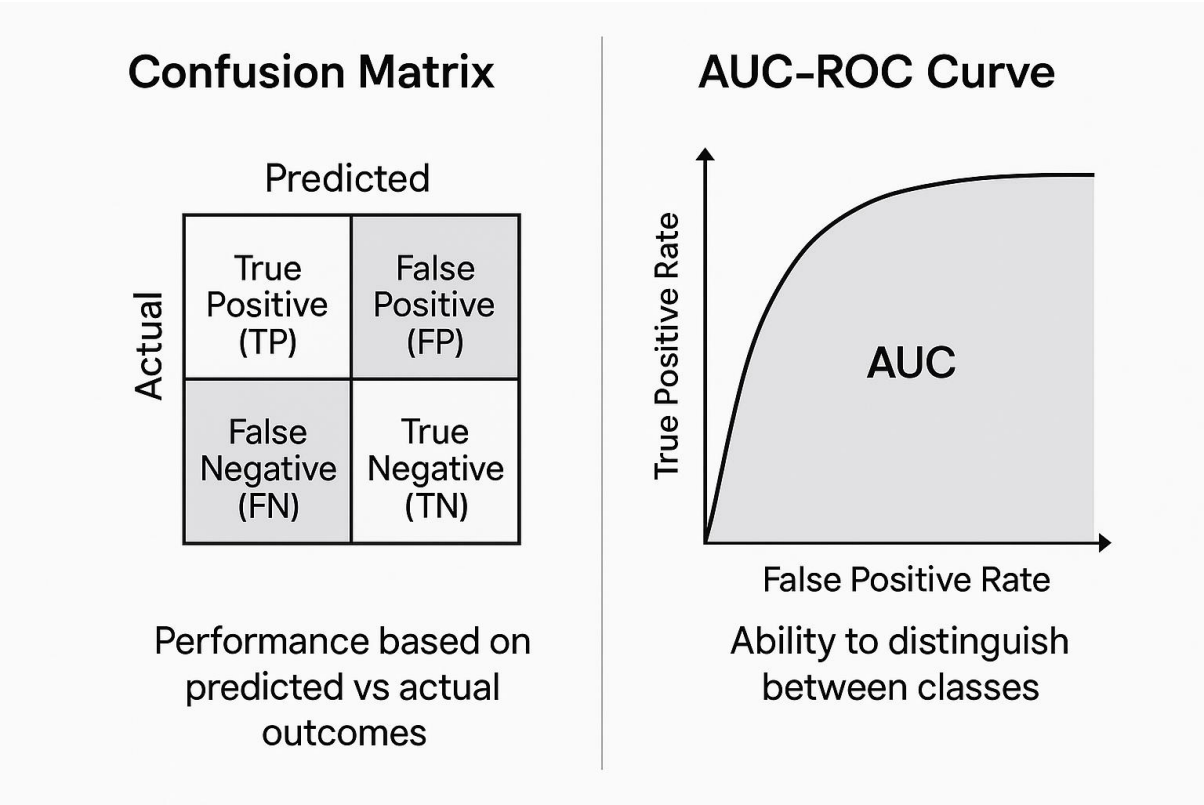
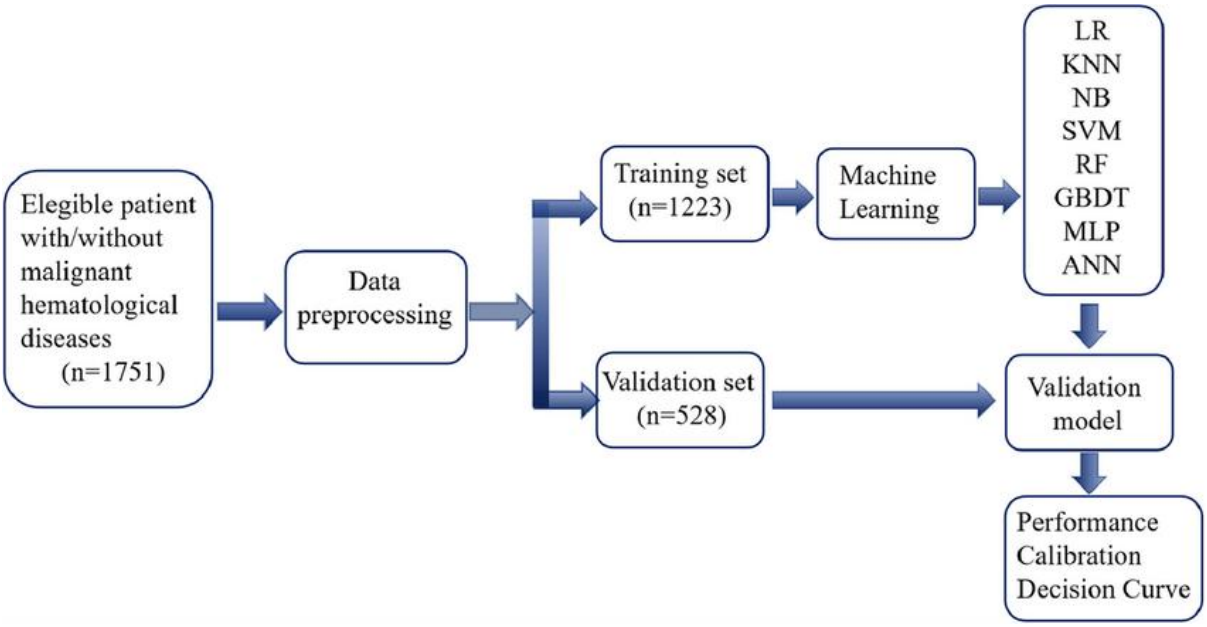
● **Detailed Flowchart (Step-wise System Design):**


Input Dataset
Data Preprocessing
Extract Features
Split Dataset
Training Dataset (80%)
Test Dataset (20%)
(Feature_train, label_train)
Train the Model
Feature_test
Label_test
Perform Predictions
Label_predic
Evaluate Metrics





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Detailed Flow:

Input Data → Preprocessing → EDA → Feature Selection →

→ Train (LR, KNN, SVM) → Evaluate → Compare → Final Prediction

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● **Expected Modules / Libraries to be used:**

➤ **pandas:**
Data handling, cleaning, and preprocessing

➤ **numpy:**
Numerical operations and array handling

➤ **scikit-learn:**
Provides ML models and evaluation tools

➤ **matplotlib.pyplot:**
Plotting graphs and visualization

➤ **seaborn:**
Advanced visualization (heatmaps, confusion matrix)

● **Expected Evaluation Metrics and Plots:**

➤ **Accuracy:**
Overall correctness of predictions

➤ **Precision:**
Correct positive predictions

➤ **Recall:**
Ability to detect actual positives

➤ **F1-Score:**
Balance between precision and recall

➤ **Confusion Matrix:**
Visual representation of predictions

➤ **ROC Curve & AUC:**
Model performance across thresholds

● **Advantages of the System:**

➤ Early detection of diseases
➤ Reduces manual effort of doctors
➤ Improves decision-making
➤ Scalable for large healthcare datasets
➤ Cost-effective solution

● **Limitations:**

➤ Depends on dataset quality
➤ Requires proper feature selection
➤ Model performance may vary with data

● **Required Topics from the Subject:**

➤ Machine Learning Algorithms
➤ Data Preprocessing Techniques
➤ Feature Selection Methods
➤ Classification Models
➤ Evaluation Metrics
➤ Data Visualization

● **Platform Required:**

➤ Python
➤ Jupyter Notebook
➤ Visual Studio Code

● **Books and Link Sources:**

➤ Tom M. Mitchell – *Machine Learning*

➤ Aurélien Géron – *Hands-On Machine Learning*

➤ Scikit-learn Documentation:
https://scikit-learn.org

➤ Kaggle Dataset Source:
https://www.kaggle.com